

YUBO LI

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EDUCATION

Carnegie Mellon University - <i>Ph.D. in Information Systems</i>	2022 – Present
Carnegie Mellon University - <i>M.S. in Information Systems & Management</i>	2020 – 2021
University of California, San Diego - <i>B.S. in Applied Mathematics Minor in Business</i>	2015 – 2019
<i>Leadership: Founder of the UCSD Triton Actuarial Society GPA: 3.8/4.0</i>	

PUBLICATIONS

- [Time-To-Inconsistency: A Survival Analysis of Large Language Model Robustness to Adversarial Attacks](#). ICLR 2026.
- Confidence as Control: A Survey of Confidence Utilization in Large Language Models. ACL 2026 (Under review)
- [Consistency of Large Reasoning Models Under Multi-Turn Attacks](#). ACL 2026 (Under review)
- [Skills-Bench : Benchmarking the Efficacy of Agent Skills Across Diverse Tasks](#), ICML 2026 (Under review)
- [Beyond Single-Turn: A Survey on LLM Multi-Turn Interactions](#). TMLR (Under review)
- [Firm or Fickle? Evaluating LLM Consistency Capability in Multi-Turn Interactions](#). ACL 2025.
- [No Black Box Anymore: Demystifying Clinical Predictive Modeling with Temporal-Feature Cross Attention Mechanism](#). American Medical Informatics Association 2025 Annual Symposium (3rd Place in Paper Competition).
- [Enhancing End Stage Renal Disease Outcome Prediction: A Multi-Sourced Data-Driven Approach](#). Journal of the American Medical Informatics Association, 2025.
- [Co-Alignment: Rethinking Alignment as Bidirectional Human-AI Cognitive Adaptation](#). Agents4Science 2025.
- [Towards Interpretable End-Stage Renal Disease \(ESRD\) Prediction: Utilizing Administrative Claims Data with Explainable AI Techniques](#). American Medical Informatics Association 2024 Annual Symposium.

FELLOWSHIPS

- 2025 Presidential Fellowship, Tata Consultancy Services (TCS)
- [2024 Fellowship in Generative AI in Healthcare](#), Center for Machine Learning and Health, Carnegie Mellon University
- [2023 Fellowship in Digital Health Innovation](#), Center for Machine Learning and Health, Carnegie Mellon University

RESEARCH EXPERIENCE

- **Q&A Systems for Trustworthy Organ Transplantation Guidance** **Carnegie Mellon University**
Supervisor: Ramayya Krishnan, Dean of the Heinz College | Rema Padman, CMU Professor 2025.01 - Present
 - **Q&A System**: Developed Q&A systems for organ transplantation patient care by integrating retrieval-augmented generation (RAG) with adaptive frameworks that synthesize patient-specific medical history, clinical guidelines, and dynamic risk factors to deliver tailored guidance and support.
 - **RAG/Retriever**: Engineered a hybrid retrieval pipeline combining both sparse and dense retrievers (DPR, ANCE) to accurately extract and verify medical content from over 100 transplant-specific handbooks.
 - **LLM Reasoning Framework**: Developed an incentivized reasoning mechanism that optimizes LLM performance by rewarding multi-step chain-of-question sequences, enabling the system to reason freely and proactively request critical diagnostic information.
- **LLM Alignment & Multi-Turn Consistency** **Carnegie Mellon University**
Supervisor: Ramayya Krishnan, Dean of the Heinz College | Rema Padman, CMU Professor 2024.05 - Present
 - **Distributed Training**: Architected and optimized distributed training infrastructure for LLMs (Mistral, LLaMA, Deepseek) across 4 nodes with 16 NVIDIA GH200 GPUs, implementing tensor parallelism, pipeline parallelism, and ZeRO optimizer techniques to efficiently scale model training and reduce computational overhead.
 - **Fine-tuning**: Performed fine-tuning of state-of-the-art LLMs both locally (on-premises clusters) and via OpenAI's official GPT-4o API, successfully improving model consistency, confidence-aware reasoning, and achieving state-of-the-art performance.
 - **Evaluation**: Proposed a metric to quantitatively assess multi-turn response stability, prioritizing early interaction accuracy and swift recovery from initial errors. Released a comprehensive benchmark to test the model's ability to sustain coherent and contextually accurate responses during dynamic, multi-turn conversations.
- **Chronic Kidney Disease Progression Analysis with DL & XAI** **Carnegie Mellon University**
Supervisor: Rema Padman, CMU Professor | UPMC & Highmark 2022.07 - Present
 - **Deep Learning**: Developed Temporal-Feature Cross Attention Mechanism (TFCAM), an attention-based deep learning framework that explicitly captures temporal and feature-level interactions to enhance clinical predictive modeling for Chronic Kidney Disease (CKD) progression.
 - **XAI**: Provided multi-level explainability including temporal insights, feature importance ranking, and cross-temporal feature interactions, enabling clinicians to interpret model predictions transparently and identify actionable clinical insights.

- **Enhanced Representation & Delivery:** Designed an interactive, user-friendly dashboard GUI accessible to diverse user groups; integrated LLMs to automatically generate clear, insightful summaries of predictions and model interpretations, enhancing usability and clinical decision-making support.

WORKING EXPERIENCE

- **Amazon – Applied Scientist Intern** 2025.06 – 2025.08
 - **Generative Recommendation Systems:** Developed User-LLM pipeline for personalized recommendation by pre-training custom user encoder on 172k customer interaction sequences, then fine-tuning through progressive alignment strategies (adapter-only, adapter + LLM, full fine-tuning) and pooling mechanisms.
 - Achieved superior predictive performance (60.1% Hit Ratio@3, 67.5% NDCG@3) compared to direct prompting baselines while delivering **93.4% token reduction** and **64.8% cost savings**.
 - **RAG/Search:** Architected production-ready RAG service infrastructure using FAISS HNSW indexing for 60k user embeddings to enhance recommendation contest at scale.
 - **Distributed Training:** Engineered distributed training across multi-GPUs using FSDP with mixed-precision BF16.
- **HireBeat – Data Scientist** 2019 – 2020
 - Leveraged machine learning and NLP techniques to develop a data analysis pipeline that optimized content recommendation algorithms, significantly improving personalization accuracy and user engagement.
- **R-eat (Self-employed) - Founder** 2012 – 2014
 - **Entrepreneurship:** Founded and grew an online food delivery business by enhancing customer experience through UI optimization and personalized marketing, resulting in a 60% sales increase and a loyal customer base of over 2,000.

COURSES

- **Guest Lecture**
 - GenAI in Healthcare | 94-706 Healthcare Information Systems | Spring 26
 - AI Agents | 95-891 Introduction to AI | Fall 25
 - Deep Learning for NLP | 95-891 Introduction to AI | Spring 25
- **Teaching Assistant**
 - 11-711 Advanced NLP with Prof. Graham Neubig.
 - 10-680/10-681 Mathematical & Computational Foundation for Machine Learning with Prof. Henry Chai.
 - 95-891 Introduction to Artificial Intelligence with Prof. David Steier.
 - 95-888 Data Focused Python with Prof. Michael Simko.
 - 95-865 Unstructured Data Analysis with Prof. George Chen.
 - 95-760 Decision Making under Uncertainty with Prof. David Choi.
- **Courses**
 - 10-714 Deep Learning Systems
 - 10-715 Advanced Introduction to Machine Learning
 - 10-716 Advanced Machine Learning: Theory and Methods
 - 10-703 Deep Reinforcement Learning
 - 10-725 Convex Optimization
 - 10-805 Machine Learning with Big Data
 - 11-711 Advanced NLP
 - 11-785 Introduction to Deep Learning
 - 11-777 Multi-Model Machine Learning
 - 15-513 Introduction to Computer System
 - 15-688 Practical Data Science
 - 36-707 Regression Analysis
 - 36-731/36-732 Causal Inference

